EDUTRACK — TECHNICAL ARCHITECTURE DIAGRAM

Layered, Multi-Tenant, AI-Native Architecture for the School ERP + LXP + SIS + CRM + Finance Platform

v2 — tenant isolation · saga orchestration · AI safety layer · CDC · secrets mgmt

SECURITY & GOVERNANCE

- Encryption at Rest & In Transit
- RBAC / ABAC Access Control
- ISO 27001:2013 Controls
- Data Privacy (DPDP / GDPR)
- Immutable Audit Trails
- Secrets Mgmt (Vault / KMS)
- Vulnerability Mgmt & Pen Testing

CLIENT / EXPERIENCE LAYER

Entry points

- **Parent Mobile App** — Fees, Attendance, Homework
- **Student Mobile App** — Learning, Portfolio, Badges
- **Teacher Web App** — LMS, Gradebook, Lesson Plans
- **Admin / Principal Portal** — Dashboards, Config, Reports
- **Public Website & Landing Pages** — Admissions CRM Front-end
- **WhatsApp / IVR Chatbot Widget** — 24x7 AI Admissions Assistant
- NEW **Offline Cache & Sync Queue** — Local-first data, background sync for low-connectivity campuses

↓ requests

EDGE, IDENTITY & API GATEWAY

Perimeter & routing

- **CDN / Edge Cache** — Static assets, media
- **API Gateway** — REST / GraphQL routing
- **SSO / OpenID Connect IdP** — Auth code flow, MFA
- **WAF & DDoS Protection** — Perimeter security
- **Rate Limiting & Throttling** — Per-tenant quotas
- NEW **API Contract & Schema Registry** — Versioned OpenAPI/GraphQL schemas shared across all services
- NEW **mTLS / Device Attestation** — Mutual TLS for biometric, RFID & IoT device channels

Tenant Context Propagation: tenant_id / org_id extracted from JWT at the Gateway → carried in request headers & event metadata through every service call → enforced at the Data Layer via row-level security / per-tenant schema boundary

↓ tenant-scoped calls

CORE APPLICATION SERVICES (Microservices)

Grouped by bounded context

ACADEMIC DOMAIN

- **Academic / LMS Service** — Course Mgmt, Teaching Plans
- **Assessment & Gradebook Service** — Board-specific report cards
- **Portfolio Service** — Student e-Portfolio, HPC
- **Document & Content Service** — Certificates, Resource Library

OPERATIONS DOMAIN

- **Admissions & CRM Service** — Lead pipeline, e-Sign
- **Fee & Finance Service** — Billing, PDC, Reconciliation
- **Communication Service** — Connect: 2-way / broadcast
- **HR & Campus Ops Service** — Payroll, Transport, Hostel
- **Notification Service** — SMS / Email / Push / IVR

PLATFORM DOMAIN

- **User & Role Mgmt Service** — RBAC, Multi-tenant orgs
- **Compliance & Audit Service** — SQAAF, Academic Audit
- **Analytics & Reporting Service** — Dashboards, KPIs

Saga orchestration: cross-service transactions (e.g. a fee-overdue workflow touching Fee, Communication & Compliance services) are coordinated asynchronously over the Event/Message Bus rather than via direct service-to-service calls.

⇅ sync API calls + async events

AI / AGENTIC INTELLIGENCE LAYER

Bidirectional with Core Services

DOMAIN COPILOTS

- **Admissions Copilot** — Lead qualification & follow-up
- **Teacher Copilot** — Lesson plans & content gen
- **Student Learning Coach** — Personalized paths & revision
- **Parent Copilot** — Fee / attendance / progress Q&A
- **Finance Copilot** — Forecasting & collections
- **Principal Copilot** — Performance summaries & risks
- **Compliance Copilot** — Statutory tracking & alerts

CORE AI INFRASTRUCTURE

- **Agentic Workflow Orchestrator** — Cross-functional automation
- **LLM Gateway / Model Router** — Multi-model routing
- **RAG Engine & Vector Store** — Curriculum & policy grounding
- **ML Model Services** — Lead scoring, adaptive engine, predictive risk, plagiarism, auto-eval
- NEW **Prompt & Output Safety / Eval** — Guardrails, PII redaction & eval harness — separated from routing given DPDP obligations on minors' data

↓ read / write + streamed events

DATA & MESSAGING LAYER

Bidirectional with Core & AI Layers

- **Event / Message Bus** — Kafka: async domain events
- **Operational Databases** — Polyglot: PostgreSQL / MongoDB, per-tenant isolation
- **Cache Layer** — Redis: sessions, hot reads
- **Search Index** — Elasticsearch: content, records
- **Data Lake & Warehouse** — Analytics, ML training data
- **Object / Media Storage** — Certificates, videos, documents
- NEW **CDC Pipeline** — Debezium / Kafka Connect streams operational DB changes into the Lake & Warehouse

Retention & purge: schedules enforced per DPDP; records tied to minors are flagged for shorter retention windows and prioritized deletion on request.

↓ outbound sync / inbound webhooks

INTEGRATION LAYER (External Systems)

Third-party & regulatory

- **Payment Gateways & eNACH** — UPI, Cards, Net Banking
- **NBFC / Bank Loan Partners** — Student & School Financing
- **Accounting ERPs** — Tally, Zoho, QuickBooks, SAP
- **Communication Channels** — WhatsApp, SMS, Email, Teams, Meet
- **Content Sources** — OER, YouTube, AR/VR, STEM Labs
- **Regulatory / Board Systems** — CBSE, State Boards, UDISE+
- **Biometric / RFID / IoT** — Attendance, Transport GPS
- NEW **Webhook / Callback Receiver** — Validates async inbound callbacks from payment & NBFC partners, queues onto Event Bus

↓ provisioned on

CLOUD INFRASTRUCTURE & PLATFORM

Foundation

- **Kubernetes / Container Orchestration**
- **CI/CD Pipeline** — GitOps, automated testing
- **Multi-Tenant Isolation** — Per-org data & config boundary
- **Auto-Scaling & Load Balancing**
- **Backup & Disaster Recovery** — RPO/RTO managed
- NEW **Secrets Management** — Vault / KMS — centralized credential & key issuance
- NEW **Environment Promotion Path** — Dev → Staging → Prod, gated by automated tests + manual approval for Prod

OBSERVABILITY & DEVOPS

- Centralized Logging (ELK)
- Metrics & Dashboards
- Distributed Tracing
- Alerting & Incident Mgmt
- SLA / Uptime Monitoring
- Feature Flags & Config Mgmt

Note: Arrows indicate primary request/data flow (client → edge → services → data); the ⇅ markers show that the AI Layer and Data Layer interact **bidirectionally** with Core Application Services (synchronous API calls plus asynchronous events via the Message Bus), not in a single downstream pass. Security & Governance and Observability & DevOps are cross-cutting concerns applied across every layer, not a single-point stage. Core Application Services are grouped into Academic, Operations and Platform bounded contexts to reflect team ownership and data-store boundaries. **New in v2:** explicit tenant-context propagation, API schema registry, saga-based cross-service orchestration, a dedicated AI safety/eval component, CDC pipeline into the data lake, inbound webhook handling, secrets management, and an explicit environment promotion path.

RBAC – Role-Based Access Control; ABAC – Attribute-Based Access Control; DPDP – Digital Personal Data Protection Act; RPO/RTO – Recovery Point/Time Objective; ELK – Elasticsearch, Logstash, Kibana; UDISE+ – Unified District Information System for Education Plus (India); CDC – Change Data Capture.